

MACHINE LEARNING PROJECT REPORT

Breast Cancer Diagnosis

Classification Using Support Vector Machines (SVC)

Wisconsin Diagnostic Breast Cancer (WDBC) Dataset

Kernel-Based Classification · Hyperparameter Optimization · Model Evaluation

PREPARED BY

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MODEL

Support Vector Classifier

1 Executive Summary

This report documents the end-to-end development of a Support Vector Classification (SVC) model to distinguish malignant from benign breast tumors using the Wisconsin Diagnostic Breast Cancer (WDBC) dataset. The workflow spans data cleaning, percentile-based outlier capping, correlation analysis, an 80/20 stratified train-test split, exhaustive hyperparameter tuning via 5-fold cross-validated grid search, and a full evaluation suite including confusion matrix, ROC-AUC, and a 2D principal-component decision-boundary visualization.

The tuned RBF-kernel SVC achieved 97.37% test accuracy with a 0.99 ROC-AUC, correctly classifying 111 of 114 held-out cases.

97.37% Test Accuracy	97.58% CV Accuracy (5-fold)	0.99 ROC-AUC	110 Support Vectors
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2 Dataset Overview

The dataset contains 569 patient records described by 30 numeric features computed from digitized images of fine needle aspirate (FNA) biopsies of breast masses. Each feature captures a statistical measure — mean, standard error, or "worst" (largest) value — of a cell nucleus characteristic such as radius, texture, perimeter, area, smoothness, compactness, concavity, and symmetry.

Attribute	Value
Total Records	569
Original Columns	33 (id, diagnosis, 30 features, 1 unnamed artifact column)
Feature Columns After Cleaning	30
Target Variable	diagnosis (M = Malignant → 1, B = Benign → 0)
Duplicate Rows	0
Missing Values	0

Class Distribution

The target variable is moderately imbalanced, which motivated the use of a stratified train-test split to preserve class proportions in both partitions.

Class	Count Share	Description
Benign (0)	62.7%	Non-cancerous tissue mass
Malignant (1)	37.3%	Cancerous tissue mass

3 Data Cleaning & Preprocessing

3.1 Column Pruning

Two non-predictive columns were removed prior to modeling:

- `id` — a unique patient identifier with no diagnostic value.
- `Unnamed: 32` — an empty artifact column introduced during CSV export.

Post-cleaning shape: 569 rows × 31 columns (30 features + target).

3.2 Outlier Detection & Capping

Outliers were identified using the 1st and 99th percentile thresholds for every numeric feature. Features such as `radius`, `texture`, `perimeter`, `area`, `smoothness`, `compactness`, and `symmetry` (mean, SE, and worst variants) each showed 12 boundary outliers, while the `concavity`- and `concave-points`-related features showed 6. All flagged values were capped (winsorized) to their respective percentile bounds rather than removed, preserving sample size while limiting the influence of extreme measurements.

Outlier Treatment Method

- Technique: Percentile-based winsorization (1st / 99th percentile clipping)
- Applied to: All 30 numeric feature columns
- Verification: Post-capping check confirmed 0 remaining out-of-bound values

4 Feature Correlation Analysis

After encoding the target (`Malignant` = 1, `Benign` = 0), Pearson correlation was computed between each feature and the diagnosis label. The five most strongly correlated features — all describing tumor size and shape irregularity — are shown below, indicating that larger, more irregular masses are strongly associated with malignancy.

Rank	Feature	Correlation with Diagnosis
1	<code>concave points_worst</code>	0.795
2	<code>perimeter_worst</code>	0.790
3	<code>concave points_mean</code>	0.785
4	<code>radius_worst</code>	0.783
5	<code>area_worst</code>	0.751

5 Train-Test Split & Feature Scaling

The cleaned feature set was split into training (80%) and test (20%) partitions using stratified sampling to preserve the benign/malignant ratio in both sets. Features were then standardized using `StandardScaler`, fit exclusively on the training partition to prevent data leakage into the test set.

Split	Samples	Feature Shape
Training Set	455	(455, 30)
Test Set	114	(114, 30)

6 Hyperparameter Tuning

An exhaustive grid search with 5-fold cross-validation was performed over kernel type, regularization strength (C), and kernel coefficient (gamma) to identify the optimal SVC configuration.

Hyperparameter	Values Searched
kernel	linear, rbf
C	0.1, 1, 10, 100
gamma	scale, 0.1, 0.01, 0.001

Best Configuration Found

- Kernel: RBF (Radial Basis Function)
- C (regularization): 1
- Gamma: scale
- Best cross-validated accuracy: 97.58%

7 Model Evaluation on Test Set

The tuned SVC was evaluated on the untouched 114-sample test partition, achieving 97.37% accuracy.

7.1 Confusion Matrix

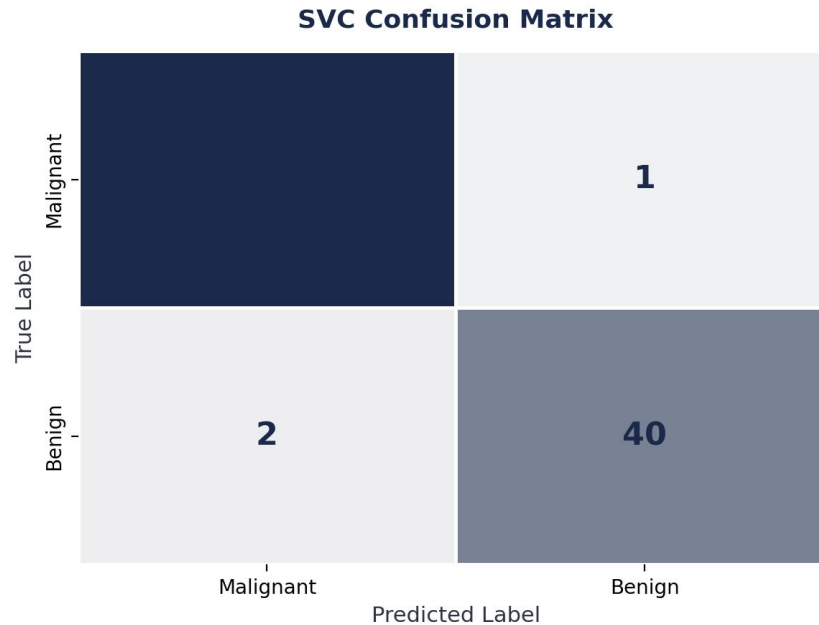


Figure 1 — Confusion matrix on the held-out test set (71 True Malignant, 40 True Benign, 3 misclassifications).

7.2 Classification Report

Class	Precision	Recall	F1-Score	Support
Malignant	0.97	0.99	0.98	72
Benign	0.98	0.95	0.96	42
Accuracy			0.97	114
Macro Avg	0.97	0.97	0.97	114
Weighted Avg	0.97	0.97	0.97	114

7.3 ROC Curve

The ROC curve illustrates the trade-off between true positive rate and false positive rate across classification thresholds. The model achieves an AUC of approximately 0.99, indicating near-perfect separability between the two classes.

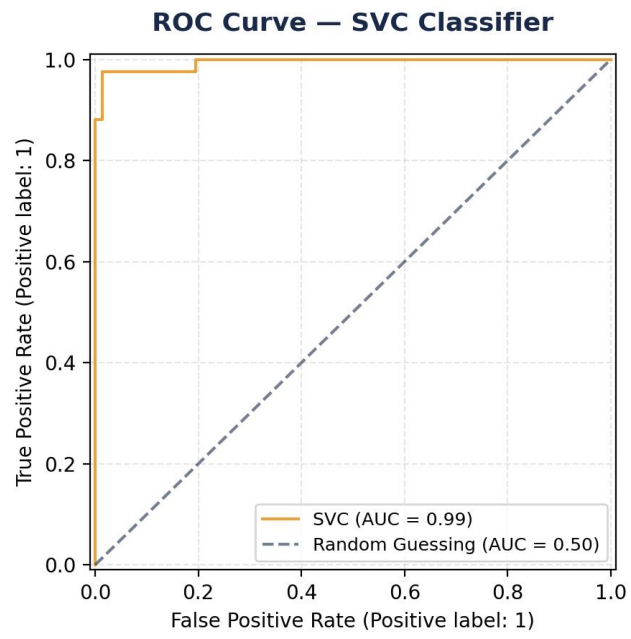


Figure 2 — ROC curve for the tuned RBF-kernel SVC, well above the random-guessing diagonal.

8 Support Vector Analysis

Support vectors are the training samples that lie closest to the decision boundary and directly determine its shape. Of the 455 training samples, 110 (24.2%) were retained as support vectors.

Metric	Value
Support Vectors — Malignant class	53
Support Vectors — Benign class	57
Total Support Vectors	110
Support Vector Feature Shape	(110, 30)
Dual Coefficient Shape	(1, 110)

9 2D Decision Boundary Visualization

To visualize how the SVC separates the two classes, the 30-dimensional feature space was projected onto its first two principal components and a fresh SVC (same tuned hyperparameters) was fit on this 2D representation purely for illustration. The shaded regions show the model's predicted class for any point in the reduced space; circled markers indicate the resulting 2D support vectors.

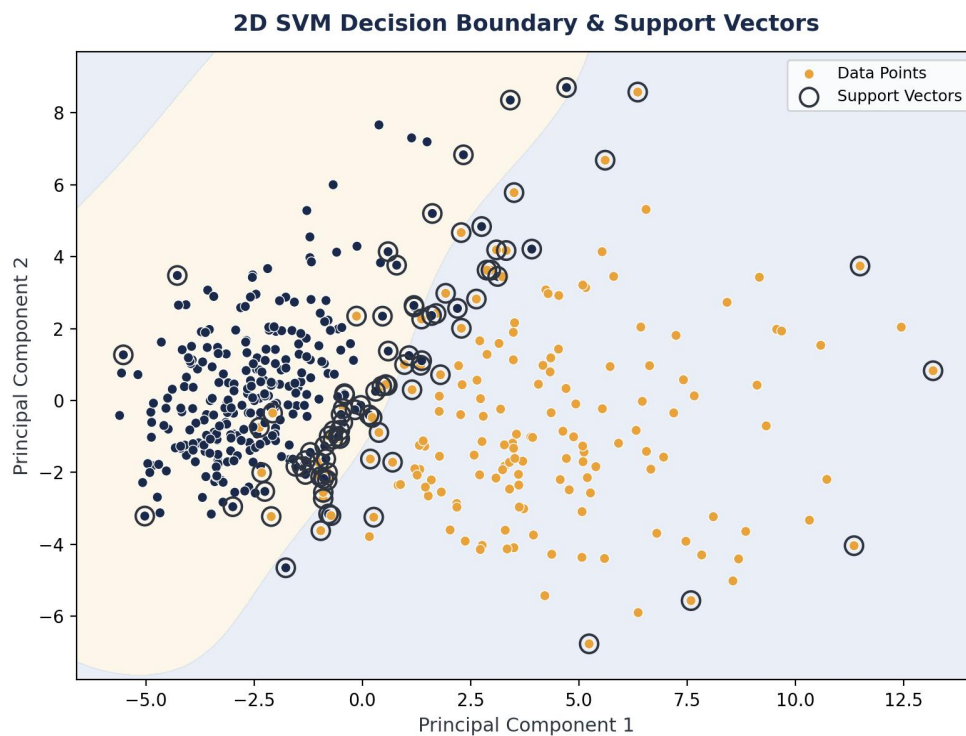


Figure 3 — PCA-reduced (2D) decision boundary with support vectors highlighted.

The two classes form largely separable clusters even in just two dimensions, with support vectors concentrated along the boundary region — consistent with the high accuracy achieved in the full 30-dimensional feature space.

10 Key Insights

- Tumor size and shape irregularity (worst-case radius, perimeter, area, and concave points) are the strongest predictors of malignancy.
- An RBF kernel outperformed a linear kernel, indicating the classes are not perfectly linearly separable in the original 30-dimensional space.
- Moderate regularization ($C = 1$) with the default "scale" gamma generalized best, avoiding both under- and over-fitting.
- The model shows higher recall for malignant cases (0.99) than benign (0.95) — clinically favorable, since missing a malignant case is far costlier than a false alarm.
- Only 3 of 114 test cases were misclassified: 1 malignant case predicted benign, and 2 benign cases predicted malignant.

11 Recommendations & Next Steps

- Investigate the single false-negative (malignant → benign) case in detail, as false negatives carry the highest clinical risk.
- Test probability-calibrated SVC (probability=True) to enable threshold tuning that further reduces false negatives.
- Benchmark against ensemble methods (Random Forest, Gradient Boosting) to validate whether the RBF-SVC's performance generalizes best.
- Apply k-fold cross-validation on the full dataset (not just the training split) for a more robust generalization estimate.
- Consider feature selection using the top-correlated features to build a lighter, more interpretable clinical decision-support variant.

End of Report